

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 6: SHAPE MODELS AND SHAPE RECOGNITION

SLO 2: Apply shape recognition techniques to classify or identify objects based on their geometric features.

SHAPE FEATURE IDENTIFICATION SOLUTIONS

Area	Aspect Ratio	Boundary Shape	Identified Shape
Medium	1 : 1	Smooth curve	Circle / Sphere
Large	2 : 1	Straight edges	Rectangle
Small	1 : 3	Long and narrow	Line / Rod / Stick

SHAPE CLASSIFICATION SOLUTIONS

Object	Area (pixels)	Aspect Ratio	Shape Hint	Classification
A	50	1 : 1	Rounded boundary	Circle (or Disc)
B	80	3 : 1	Long rectangular shape	Rectangle (or Elongated Bar)
C	40	1 : 1	Straight edges	Square

ACTIVITY 3 – FEATURE MATCHING SOLUTIONS

Shape Feature	Correct Match / Purpose
Area	B (Determines object size)
Perimeter	A (Measures object boundary length)
Aspect Ratio	C (Identifies elongation of shape)
Compactness	D (Measures how circular an object is)